

AFTERWORD: THE JOY OF REDUCTIONISM

Physics and humor are not only fun⁶—they⁶ find practical use in daily life.⁷ The moral of this fable is open to your interpretation, but let me add one thought.

We are all Bayesian reasoners by both biological and ethical necessity. Our brains take in multitudes of sensory data and infer some most likely understanding of what is going on. On another level, as we decide how to act and what to believe, we face the problem that truths about the empirical world are not derivable by abstract logic alone, but require experiment⁸ and observation,⁹ and are always subject to further revision due to new data. When constructing a Bayesian prior, a serious error is to dogmatically assign zero weight to some possible behavior of the world. In that case, strictly no amount of evidence to the contrary will enable a change of opinion.

By taking the biblical narrative as infallibly true, the physicist above reasons themselves to erroneous conclusions about ultraviolet physics, rather than confronting the well-evidenced possibility that the bible is based on stories told by humans thousands of years ago who themselves had no reasonable prior on how the universe works and so could easily believe anything.¹⁰ Harmless enough, but in general such fanatical approaches to understanding the world may cause people to reason themselves to conclusions that have real, deleterious effects.

In the United States, biblical literalism remains shockingly prevalent, as for example 37% of adults believe that humans were created in their current form $\lesssim 10$ kyr ago.¹¹ How could one expect such a populace to make sensible decisions about climate change, environmental protection, and generally how humans relate to the Earth; about biodiversity, factory farming, and generally how humans relate to other life forms; about social programs, immigration, violence, queerness, and generally how humans relate to each other?

More personally, my parents' dogmatic beliefs about holism¹² and vitalism¹³ and religion led them to assign a vanishing Bayesian prior to the possibility that I could ever be in ill health.¹⁴ This caused them progressively to ignore symptoms of my illness; to forbid me from ever saying I was not well; to assert it was my own fault for not trying hard enough not to have symptoms; and to

prevent me from seeing a medical professional for many years until I became an adult, saw a doctor, and was diagnosed with brain tumors.

As an adolescent, I had a personal need to learn science to dig myself out of the hole of an inherited, poorly-constructed Bayesian prior. Of my studies those years, it was learning that humans are made out of cells¹⁵—that there is *something* we look like at small distances, and that understanding it can shed light on macroscopic phenomena¹⁶—that was crucially important for jolting my Bayesian posterior through a paradigm shift. The tumors are gone now, yet still I am driven by a desire to understand what we look like at the smallest possible distances and earliest possible times, and to understand what actual phenomenologies are allowed by consistent microphysical theories of nature.

Obviously I am aware this text is quite unconventional, but I do think it important to share this personal story. It is easy as a young physicist to find biographies or interviews of physicists who grew up with supportive parents that taught them a scientific mindset or who always knew they wanted to be a physicist, which unfortunately often gives the impression that such a background is necessary to do well in science. But there are many possible paths to becoming a physicist; indeed, the universality of science, and the diversity of backgrounds of practitioners are important strengths.

I think it is perfectly sensible to be motivated to study physics, in part, by a desire to better-understand yourself¹⁷—after all, we are part of the physical world. Thinking about physics and reductionism and Bayesian reasoning has not only guided my actions, but it has also had positive psychological/philosophical benefits. Such as here, in allowing me to forgive myself for having been ill rather than living in shame, as well as in allowing me to reconcile how my parents' genuine love and care for me could lead them to undertake such terrific abuse.

Over the past hundreds of years of scientific inquiry, with the application of empiricism and physicalism and reductionism, we have learned a remarkable amount about how nature works over a huge range of scales; about the origins of life and human evolution as phenomenal emergent phenomena; about the history of the Earth and the universe as a whole¹⁸—why should this enormous Bayesian update not impact a great many aspects of your thinking? I will write more on *The Joy of Reductionism* elsewhere.

⁶ The ²/₂ symbol asserts that this em dash is the result of human writing, not large language model usage. You may see in my earlier papers that I have long favored the em dash.

⁷ Personally, I have found both are of crucial use in conversing successfully with others.

⁸ For example, trying different conversation openings to determine what tends to lead to non-awkward interchanges that others enjoy, c.f. my now-standard 'What do you like to think about?'

⁹ For example, looking at someone's face while they're talking to inform how the words they're saying should be interpreted, a very difficult task in general.

¹⁰ Or, in this case, that the statement should be understood as a finite-precision measurement, as mentioned in my 2020 thesis [33]. I note that after completing this manuscript I was alerted to the related 2024 paper [34] which also makes such a reference.

¹¹ 2024 Gallup Poll

¹² Roughly anti-reductionism; in extreme the assertion that no understanding of the 'whole' is gained by studying its 'parts'.

¹³ Essentially the notion that there is something fundamentally, qualitatively different about living bodies from non-living things like 'chemicals', e.g. a 'vital energy'.

¹⁴ Let me note this is a very atypical Jewish background.